Answer key Tutors only

The numbers the web pages show. ADC in × 10^{-3} mm²/s. "Say" = the one point to make at the debrief. Block order B, C, A (Intro 10 · B 25 · C 25 · break 10 · A 30 · debrief 15 min).

B Has the tumour responded? · first

Patient PCAMPMRI-00011, shown in reverse order

2 **Scan 1 `1.192` · Scan 2 `1.310` · Change `+10` %** (volume 0.55 → 0.39 cm³, −28 %)

3

ROI on scan 1	Mean	Median	10th percentile	Change, scan 1 → 2 (mean)
Expert, on ADC	1.192	1.187	0.984	+10 %
Grown 2 mm	1.283	1.248	1.011	+4 %
Shrunk 2 mm	1.119	1.142	0.983	+14 %
Expert, on T2w	1.503	1.560	1.192	**−11 %**

Moves least: the 10th percentile. It stays at 0.98–1.01 on the three ADC outlines, while the mean runs from 1.12 to 1.28. The T2w outline shifts every value, because on the ADC map it covers partly different tissue.

Say: The ROI alone moves the mean by −6 % to +26 % on one scan. Low percentiles are less sensitive to the edge.

4 **Cannot tell. +10 % is far below the prostate tumour limit (42 %, Fedorov 2017; 47 %, QIBA). −28 % volume is well within the volume limit (112 % on ADC).**

> **Reveal:** there was no treatment. The two scans are a test-retest pair 13 days apart, shown in reverse. In the order they were taken, ADC *fell* 9 %. Across the 10 men in the study, ADC changed between −9 % and +43 % with no treatment.

5 **Yes. Outline drawn on ADC: +10 %. The same radiologist's outline drawn on T2w: −11 %. Opposite directions.**

Bonus

C Contouring on DWI · second

Patient PCAMPMRI-00007, first scan · 3 mm slices

1–3 **Pairs' own numbers. For comparison, the radiologist: 0.38 cm³ on T2w, 0.37 cm³ on ADC. The ruler should show a few mm at the back of the prostate (the radiologist's two outlines: 2.8 mm front-back).**

Say: Similar volumes do not mean the same place. Look at the overlap and the shift, not only the size.

Bonus **Behind the blue line, towards the balloon: about 5 mm on the middle slices, up to about 10 mm in places. The front edge stays close to the line.**

Say: The whole prostate is pushed towards the air, not only the tumour. An outline drawn on DWI lands several mm too far back on the planning images.

4 **Radiologist: Dice `0.23` · DWI outline `2.8 mm` further back (`5.1 mm` between the centres in all, partly towards the feet).**

Say: If you can't find the T2w outline, check the slice: T2w is drawn on slices 8–11, ADC on slices 7–9.

DISCUSS (NO SINGLE RIGHT ANSWER)

1 **Most centres: a guide, or a boost volume after checking the geometry. In FLAME the boost was contoured on multiparametric MRI (T2w, DWI and DCE together), not on DWI alone.**

2 **At least the shift measured here: a few mm (5 mm between the radiologist's outlines). It changes from patient to patient, so it can't be a fixed number.**

3 **Rigid registration does not fix it: the distortion is local and non-linear. Distortion correction helps (reverse phase-encoding scan or a B0 field map). Check the prostate edge on b 0 against T2w, especially near the rectal air or balloon.**

Prostate data: QIN-PROSTATE-Repeatability, TCIA, doi:10.7937/K9/TCIA.2018.MR1CKGND (CC BY 4.0); Fedorov et al., Sci Data 2018;5:180281 and Invest Radiol 2017;52:538-46. FLAME: Kerkmeijer et al., J Clin Oncol 2021;39:787-96.

TAB 1 · SAME BRAIN, DIFFERENT ADC

1 **CSF (ventricles, cisterns) and the vessels in the Sylvian fissure: about a third of their signal goes between b 0 and b 100. Brain tissue loses about 9 %.**

Say: At low b-values, moving blood and CSF dominate the signal. That is not the diffusion you want to measure.

2 **Brain tissue: b 0 & 800 `0.79` · b 200 & 800 `0.75`**

3 **ADC (b 0 & 800) vs ADC (b 200 & 800), the Difference map: vessels + CSF (Sylvian fissure) +69 % · brain +5.2 % · CSF +5.8 %**

Say: More blood flow in the region means a bigger jump when b 0 is in the fit. Most pairs guess CSF.

4 **Scan 1 b 0 & 800 (`0.79`), follow-up b 200 & 800 (`0.75`): the ADC "falls" about 5 % although the tissue did not change. That is about half of the 11 % limit, used up by the protocol change alone. Both scans b 0 & 800: the offset is in both and cancels (0 %), as long as the blood flow stays the same (see Tab 2).**

Say: Same b-values on every scan. A protocol change alone can eat half of the brain limit.

TAB 2 · TREATMENT (SIMULATED: F 0.25 → 0.08; D = 1.0 AND D* = 20 DO NOT CHANGE)

1 **Guess: with b 0 the ADC falls; with b 200 it stays the same. Real values:**

	Before	After	Change
b 0 & 800	1.36	1.10	**−19 %**
b 200 & 800	1.01	1.00	−0.9 %
b 50 & 800 (task 4)	1.16	1.04	−10 %

2–3 **Depends on the b-values. With b 0 in the fit, the ADC falls 19 %, more than the 11 % QIBA limit for brain, so it counts as a real change. A tumour that responds usually shows a *rising* ADC (dying cells leave water freer to move), so this fall would be read as no response. With b 200 & 800 the ADC moves 0.9 %.**

Say: The cells never changed, only the blood did. The b-values decided the verdict.

4 Bonus **No. b 50 halves the effect (−10 %) but does not remove it, and −10 % is just under the 11 % limit. The Unity MR-linac consortium advises a lowest b of 100-150 (Kooreman 2020).**

TAB 3 · NOISE (tasks 3-5 are bonus since 2026-11-10)

1–2 **Clear: D. Noise: f and D* (D* is the worst).**

3 Bonus **f is high, usually 0.5-1.0 (median 0.9), different in each voxel. It makes no sense: CSF has no capillary blood.**

Say: CSF's fast free diffusion looks like a "perfusion" component to the model. A fitted number is not proof that the model applies.

4–5 Bonus **SNR 50: D stays close (about 0.9-1.1, true 1.0). f wanders (0.09-0.20, true 0.15). D* ranges from 13 to 40 (true 20).**

SNR 20: only D, and even D spreads ±30 %. f runs from 0 to 0.27, D* from 4 to 150.

Say: This is why IVIM maps of one voxel are hard to trust, and why published f values disagree.

Ranges = the middle 80 % of 2,000 simulated scans. Ten scans give a similar but noisier picture.

Checklist · suggested answers

1 · Which b-values?	Lowest b at least 100-150 (not b 0), and the same b-values on every scan.
2 · Whose ROI?	The same reader and the same method each time, drawn on the ADC map. Report the median or a low percentile too.
3 · Bigger than the repeatability?	Brain 11 %, prostate tumour 42-47 %, measured on the same scanner with the same protocol.
4 · Geometrically trustworthy?	Compare DWI with T2w at the edges, near air. Correct the distortion or add a margin.

Brain data: E. Peterson, IVIM dataset, Figshare 2016, doi:10.6084/m9.figshare.3395704 (CC0). Limits: QIBA DWI Profile, Stage 3, 2022 (brain 11 %, prostate 47 %). Lowest b-value: Kooreman et al., Radiother Oncol 2020;153:106-13.